GYR-O Foundations

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Summary

GYR-O's reduction of visual clutter rests on one well-established fact: human attention and working memory are limited. Every item on a page competes for a glance and for a slot in short-term memory, so what the reader does not need still costs them something.

Eye-movement research is how that cost is measured. A cluttered page produces more fixations, longer searches and more backward jumps (regressions). GYR-O's principles map onto five bodies of research, listed below. They support the principles; they do not yet prove that GYR-O itself performs better.

Principles and their foundations

Each GYR-O principle, the research it rests on, and the cost it avoids.

GYR-O principle	Rests on	Cost avoided
First-time visitor is the highest priority	Information foraging	Leaving before finding the answer
Intro is self-contained; a brief summary is often enough	Information foraging; cognitive load theory	Reading further than needed
Intro on/off, in the visitor's hands ("ergonomics")	Cognitive load theory; visual search	Re-scanning material already known
No "hunting"	Visual search and crowding; eye movements in reading	Extra fixations and search time
Information comes to the reader	Information foraging ("information scent")	Navigation and guessing where to look
Scroll-to-text	Eye movements in reading; visual search	Locating the passage by eye
Fewer elements per screen	Hick-Hyman law; Tufte's data-ink ratio	Decision time and ink that carries nothing

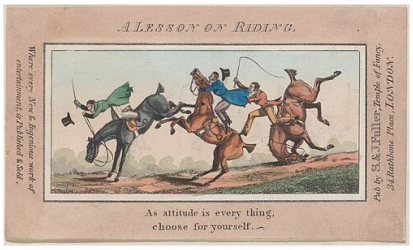
The Intro/Info switch

Two checkboxes give four states, so one page serves both the ultra-compact reader and the detailed reader without either getting in the other's way. The page works like a header and a body with no footer: Intro is the header, Info is the body.

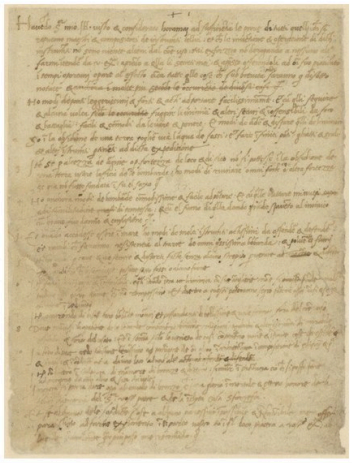
Intro	Info	What shows	Suggested name
off	off	Thumbnail of the signature graphic only	Card
on	off	Intro and the signature graphic	Handbill
off	on	Info only	Reference
on	on	Intro, signature graphic and Info	Full

"Intro" is the everyday word; what it is, in fact, is the **synopsis**. Often the synopsis is all that is necessary or desired, and anything more is visual clutter. GYR-O calls the formats built on this idea **synoptic formats**, from the Greek for "seeing all together": the whole taken in at one view. A synoptic format is concise and self-contained. A business card, a one-page resume, a circus poster, a handbill, a broadsheet and a highway billboard are all synoptic formats. Whichever one a site's Intro takes is the whole Intro, complete in itself.

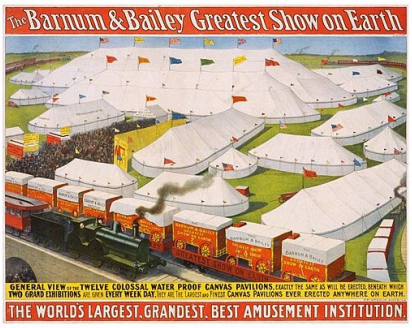
Synoptic formats: concise and self-contained, taken in at one view



Business card
Trade card, 19th c. · Met · CC0



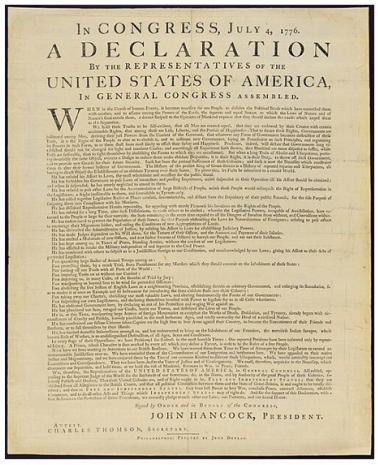
One-page resume
Leonardo to Ludovico Sforza, c.1482 · PD



Circus poster
Barnum & Bailey, 1899 · Strobridge · PD



Handbill
Mammoth Tree, 1854 · PD



Broadsheet
Dunlap broadside, 1776 · LOC · PD



Highway billboard
North Dakota, 2010 · Chitrapa · CC BY-SA 4.0

Six synoptic formats, from Wikimedia Commons: trade card (Met, CC0); Leonardo to Ludovico Sforza (public domain); Barnum & Bailey poster (public domain); Mammoth Tree handbill (public domain); Dunlap broadside (Library of Congress, public domain); highway billboard (Chitrapa, CC BY-SA 4.0).

A resume-format Intro shows how short the compact read can be. In Ladders' 2018 eye-tracking study, recruiters' initial screen of a resume averaged 7.4 seconds, up from about 6 seconds in its 2012 study. They looked first for name, current and previous titles, and dates. Resumes that did poorly had long sentences, multiple columns and little white space (Ladders 2018 PDF; HR Dive). Caveat: this was an industry study, not peer-reviewed, and the 2018 report does not give its sample size (ERE).

The design term for this is **progressive disclosure** (Nielsen): show the essentials first and let the reader open the detail. GYR-O's version is unusual because the reader can also close the essentials once they are known. That is the cognitive-load argument in the research section, put in the visitor's hands.

Open question: in the Reference state the signature graphic disappears entirely, while Card keeps a thumbnail. Keeping the thumbnail in Reference too would keep the site's identity visible in every state.

The research

Eye movements in reading. Reading is not a smooth sweep. The eye jumps (saccades, about 7 to 9 letters forward), pauses to take in text (fixations, roughly 200 to 250 ms), and sometimes jumps back to reread (regressions). Javal observed the jumps in 1879; Huey described them in 1908; Keith Rayner's work is the modern standard. These are the units in which a page's cost to the reader can be counted.

Visual search and crowding. The time to find a target grows with the number of other items on screen, and items packed close together interfere with each other (Treisman; Wolfe). This is the science behind "hunting."

Cognitive load theory. Working memory holds only a few items at once. Material the reader does not need uses capacity that understanding needs (Sweller). Letting the visitor switch Intro off removes load they have already paid.

Information foraging. Readers follow "information scent" toward what they want and give up when searching costs more than it returns (Pirolli and Card). GYR-O's "information comes to the reader" is this principle stated as design: shorten the path so the scent never has to be followed far.

Hick-Hyman law. Decision time rises with the number of choices offered. Fewer options on screen means a faster first move.

Tufte's data-ink ratio. Remove whatever on the page carries no information. Tufte's term for decoration that does no work is "chartjunk."

Grounded, not yet proven

The research shows that clutter and hunting cost the reader. It does not show that a GYR-O page costs less than a conventional one. Until that is measured, GYR-O should say its principles are *grounded in* this research, not *proven by* it.

A simple test would settle it:

1. Pick one site's content and build it two ways: a GYR-O page and a conventional page.
2. Give first-time readers the same three questions to answer from the page.
3. Time each answer and count wrong answers and give-ups.
4. If an eye tracker is available (webcam-based ones are inexpensive), also count fixations and regressions per answer.
5. Repeat with returning readers, Intro switched off, to test the on/off principle separately.

Sources

Standard references, listed from general knowledge; editions and page numbers not yet checked.

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